

Mild pH-decoupling aqueous flow battery with practical pH recovery

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Establishing a pH difference between the two electrolytes (pH decoupling) of an aqueous redox flow battery (ARFB) enables cell voltages exceeding the 1.23 V thermodynamic water-splitting window, but acid–base crossover penalizes efficiency and lifetime. Here we employ mildly acidic and mildly alkaline electrolytes to mitigate crossover, achieving high round-trip energy efficiency with open circuit voltage >1.7 V. We implemented an acid–base regeneration system to periodically restore electrolytes to their initial pH values. The combined system exhibited capacity fade rate <0.07% per day, round-trip energy efficiency >85% and approximately 99% Coulombic efficiency during stable operation for over a week. Cost analysis shows that the tolerance of acid–base crossover could be increased if the pH-decoupling ARFB achieved a higher voltage output and lower resistance. This work demonstrates principles for improving lifespan, rate capability and energy efficiency in high-voltage pH-decoupling ARFBs and pH recovery concepts applicable for pH-decoupling systems.

The need for decarbonization and the intermittency of wind and solar energy drive the quest for grid-scale storage technologies. The aqueous redox flow battery (ARFB) is a promising solution, featuring safety, the decoupling of energy and power capacities and potential long lifetime and low cost. Redox-active materials have been rapidly developed to approach the requirements for practical grid-scale storage¹.

Despite the advantages of ARFBs, the 1.23 V thermodynamic water-splitting window presents practical challenges. Competing water-splitting reactions can compromise the Coulombic efficiency of the cell² and rapidly drive the two sides of the cell out of balance, greatly cutting the operational lifetime³. Operation of ARFBs at higher voltage is a potential pathway to achieving increased energy density, higher output power and lower capital cost, all of which improve the techno-economy and operational flexibility of the cell⁴. Efforts have been put into designing redox-active species that allow higher cell voltage and developing conditions in which water-splitting kinetics remain sluggish^{5,6}.

Recently, pH-decoupling ARFBs, which contain negolytes (negative electrolytes) and posolytes (positive electrolytes) at different

pH, have attracted increasing attention. By employing specialized ion-selective membranes and bipolar membranes^{7,8}, single-membrane pH-decoupling ARFBs have been demonstrated. Multi-chamber cell structures have also been demonstrated to decouple the pH of negolyte and posolyte^{9–13}. For example, by using a three-chamber, two-membrane flow battery design, pH-decoupling ARFBs achieved record high voltage and power output^{10,14}. However, all previously reported pH-decoupled systems suffer from severe proton/hydroxide crossover due to the pH difference across the membrane. This results in short lifetimes and low overall energy efficiency, considering the energy cost required to rebalance the pH difference^{12,15}. Adding acid and base to restore the initial pH values is only a superficial solution as during the long-term operation for ARFBs (years), salt accumulation from the net reaction of exogenous acid and base can deteriorate the cell. Longevity of the pH-decoupled systems is an important criterion for practical deployment¹⁶, yet none of the developed pH-decoupling ARFBs has been proven to be long-term practicable.

In this work, we implemented a mild pH-decoupling design with a pH-difference recovery system to address this critical obstacle.

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Because the thermodynamic water-splitting voltage varies linearly with pH whereas crossover fluxes vary exponentially, by reducing the strength of the acidic and alkaline electrolytes, we have substantially reduced proton/hydroxide crossover while maintaining stable cycling with an open circuit voltage more than half a volt beyond the 1.23 V threshold. Even this mild pH-decoupling significantly broadens opportunities for utilizing high-voltage redox molecules. A redox pair of benzo[a]hydroxyphenazine-7/8-carboxylic acid¹⁷ (BHPC) and tris(4,4'-bis(hydroxymethyl)-2,2'-bipyridine)iron dichloride¹⁸ (Fe(Bhmbpy)₃) that function in mild alkaline and acidic conditions, respectively, were chosen to demonstrate the performance of this mild pH-decoupling design. To compensate for remaining acid–base crossover, the pH difference was intermittently recovered using an acid–base regeneration cell, which does not require exogenous acid or base and thereby avoids salt accumulation. Because of the low observed crossover flux, the energy cost and capital cost of this pH recovery process is negligible. We achieved a stable high-voltage ARFB with a fade rate <0.07% per day (<0.002% per cycle), round-trip energy efficiency >85% (pH recovery energy cost included), Coulombic efficiency around 99% and peak power density of 140 mW cm⁻². The cell was cycled continuously for over one week while maintaining its pH difference with pH recovery, demonstrating the suitability of the design for practical long-term operation. This approach has resulted in one of the most stable high-voltage ARFB performances to date. We further demonstrate other pH-decoupling cell structures and different aqueous redox chemicals utilizing the concept of mild pH-decoupling and pH-difference recovery, guiding the design of pH-sensitive or pH-unstable systems.

Acid–base crossover in pH-decoupling systems

Our design, using a three-chamber two-membrane cell structure with an acid–base generator, is illustrated in Fig. 1a. The three-chamber two-membrane cell was selected as the model system due to the transparency it provided for the study of proton/hydroxide crossover. We selected Fe(Bhmbpy)₃ at pH 2–4 with a supporting electrolyte of NaCl/HCl for the posolyte and BHPC at pH around 13 with a supporting electrolyte of NaCl/NaOH for the negolyte. The open circuit voltage (OCV) of this cell was approximately 1.75 V. An anion exchange membrane (AEM) and a cation exchange membrane (CEM) were used to separate the acidic positive and alkaline negative electrolytes, respectively, from the centre chamber. NaCl solution was cycled in this centre compartment between the two membranes, and pH sensors were inserted in all three chambers' electrolyte tanks for real-time pH monitoring. To counteract any remaining proton/hydroxide crossover processes, the posolyte and negolyte tanks were also connected to a bipolar membrane sub-cell (BPM sub-cell) for periodic pH recovery. During its operation, the BPM sub-cell reduces the posolyte and oxidizes the negolyte, dissociating water into protons and hydroxide, generating acid and base for pH-difference recovery. This BPM sub-cell can be small, relative to the main cell, and operate at intervals for periodic pH recovery. The sub-cell can also operate at steady state under constant current for operando pH recovery. Because operando pH recovery can maintain the electrolyte pH as constant, we chose the periodic pH recovery to show the change during cell cycling.

To achieve ion selectivity, we utilized AEMs and CEMs to separate the acidic and alkaline sides, respectively. In this system, sodium cations and chloride anions serve as the major charge carriers. The main charge carrier concentration is almost identical across the liquid junctions of the two membranes and thus the battery open circuit voltage is directly determined by the difference between the two electrode potentials. Although hydroxide and protons are mainly excluded by the CEM and AEM, respectively, the pH gradient that exists in the cell nevertheless causes undesired crossover of these ions¹⁹.

The cross-sectional ion concentration profile and the resulting transport expected during operation of the cell are shown in Fig. 1b.

Most hydroxide ions are blocked by the CEM. However, small amounts of hydroxide ions can cross and increase the centre chamber pH before eventually exchanging through the AEM. The symmetric process can occur for protons. The crossover rate of protons and hydroxide can be described by equations (1) and (2). $[H^+]_A$, $[H^+]_C$ and $[H^+]_B$ represent the proton concentrations in acidic, centre and basic chambers, respectively; their chemical compositions are assumed to be spatially uniform. $[OH^-]_A$, $[OH^-]_C$ and $[OH^-]_B$ represent the hydroxide concentrations in acidic, centre and basic chambers. A_{mem} represents the membrane area, V_A represents acidic chamber volume and l_{AEM} represents the thickness of AEM. j represents the signed ionic current, with direction pointing from acidic to alkaline chamber as positive. P represents the permeability coefficient of ions through the membrane. M represents the ion-migration coefficient inside the membrane due to the applied electric field. Specifically, the ion-migration coefficients can be expressed as equations (3)–(6). Here D represents the diffusivity of an ion in a membrane. Z is the signed charge number of the ion. F is Faraday's constant. R is the universal gas constant and T is the absolute temperature. σ is conductivity of the membrane in the electrolyte.

$$-\frac{d[H^+]_A}{dt} = \frac{A_{mem}}{V_A} \left(P_{H^+}^{AEM} \frac{[H^+]_A - [H^+]_C}{l_{AEM}} + M_{H^+}^{AEM} j + P_{OH^-}^{AEM} \frac{[OH^-]_C - [OH^-]_A}{l_{AEM}} + M_{OH^-}^{AEM} j \right) \quad (1)$$

$$-\frac{d[OH^-]_B}{dt} = \frac{A_{mem}}{V_B} \left(P_{OH^-}^{CEM} \frac{[OH^-]_B - [OH^-]_C}{l_{CEM}} + M_{OH^-}^{CEM} j + P_{H^+}^{CEM} \frac{[H^+]_C - [H^+]_B}{l_{CEM}} + M_{H^+}^{CEM} j \right) \quad (2)$$

$$M_{H^+}^{AEM} = D_{H^+}^{AEM} [H^+]_{AEM} \frac{ZF}{\sigma_{AEM} RT} \quad (3)$$

$$M_{OH^-}^{AEM} = D_{OH^-}^{AEM} [OH^-]_{AEM} \frac{ZF}{\sigma_{AEM} RT} \quad (4)$$

$$M_{OH^-}^{CEM} = D_{OH^-}^{CEM} [OH^-]_{CEM} \frac{ZF}{\sigma_{CEM} RT} \quad (5)$$

$$M_{H^+}^{CEM} = D_{H^+}^{CEM} [H^+]_{CEM} \frac{ZF}{\sigma_{CEM} RT} \quad (6)$$

$$J_{crossover} = \max \left(-\frac{d[H^+]_A}{dt} \frac{V_A}{A_{mem}}, -\frac{d[OH^-]_B}{dt} \frac{V_B}{A_{mem}} \right) \quad (7)$$

As shown in Fig. 1b, we refer to $P_{OH^-}^{CEM} \frac{[OH^-]_B - [OH^-]_C}{l_{CEM}} + M_{OH^-}^{CEM} j$ as the hydroxide crossover terms (through the CEM) and $P_{H^+}^{AEM} \frac{[H^+]_A - [H^+]_C}{l_{AEM}} + M_{H^+}^{AEM} j$ as the proton crossover terms (through the AEM). Similarly, we refer to $P_{OH^-}^{AEM} \frac{[OH^-]_C - [OH^-]_A}{l_{AEM}} + M_{OH^-}^{AEM} j$ as the hydroxide exchange terms (through the AEM) and $P_{H^+}^{CEM} \frac{[H^+]_C - [H^+]_B}{l_{CEM}} + M_{H^+}^{CEM} j$ as the proton exchange terms (through the CEM). The net acid–base crossover flux ($J_{crossover}$) of the system can be described by equation (7). The equivalent acid–base crossover current density can be calculated by multiplying net acid–base crossover flux with Faraday's constant.

Initially, the centre-chamber NaCl solution is at near-neutral pH. In our specific system, hydroxide crossover flux from the alkaline negolyte (pH 13) is larger than proton crossover flux from the acidic posolyte (about pH 3). Thus, over time, the pH of the centre chamber increases. Higher centre pH increases the hydroxide exchange rate through the AEM, and any exchanged hydroxide ion entering the acidic chamber is neutralized by a proton. Thus, the net effect of hydroxide exchanged

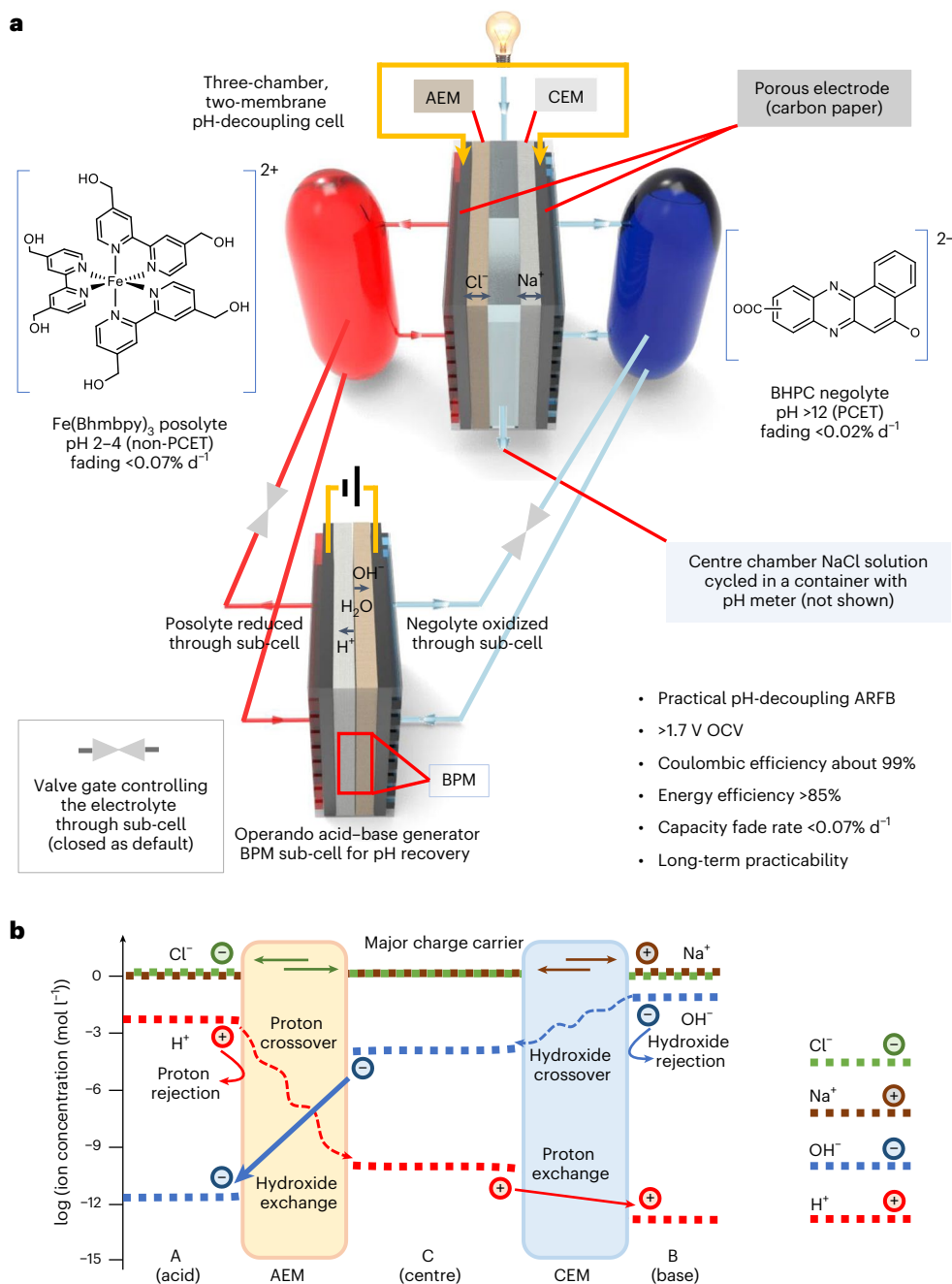


Fig. 1 | Cell structure and charge carrier migration. a, Schematic of a three-chamber two-membrane cell with a pH recovery sub-cell. Electrolytes, cell components and sensors are labelled accordingly. Electrolyte flow is indicated

with arrows. Electrolytes are pumped into the BPM sub-cell for acid–base regeneration when needed (pumps not shown). **b**, Ion concentration and flow across three chambers. Sodium and chloride ions are the major charge carriers.

across the AEM is the loss of protons in acidic chamber. Eventually, the centre chamber pH will reach steady state, at which point the net flux of protons and the net flux of hydroxide into the centre chamber are equal.

A large amount of crossover will destroy the designed pH gradient by driving all chambers towards identical pH values, resulting in diminished operating voltage and possibly instability of the redox-active molecules.

To characterize the hydroxide and proton crossover flux independently for each membrane under alternating electric field during charging and discharging, we constructed a two-chamber supercapacitor cell, with one chamber having an alkaline (or acidic) initial state as crossover source and the other chamber having a near-neutral pH as

acceptor chamber (Fig. 2a). The two sides were supported by NaCl solutions and separated by a CEM (or AEM) for hydroxide (or proton) crossover testing. Supercapacitors can be charged and discharged through ion adsorption, without introducing redox-active molecules that could interfere with the measurements. Ideally, only Na⁺ (or Cl⁻) ions should cross the membrane when the cell is subjected to an alternating current. However, ion-selective membranes cannot perfectly reject hydroxide ions and protons, leading to a pH change that can be observed in the initially pH-neutral acceptor chamber.

Depending on their selectivity and molecular structure, different ion-selective membranes exhibit different rates of crossover. We selected well-established and commonly used ion-selective membranes: Fumasep E-620K (E620) as the CEM and Selemion DSV-N

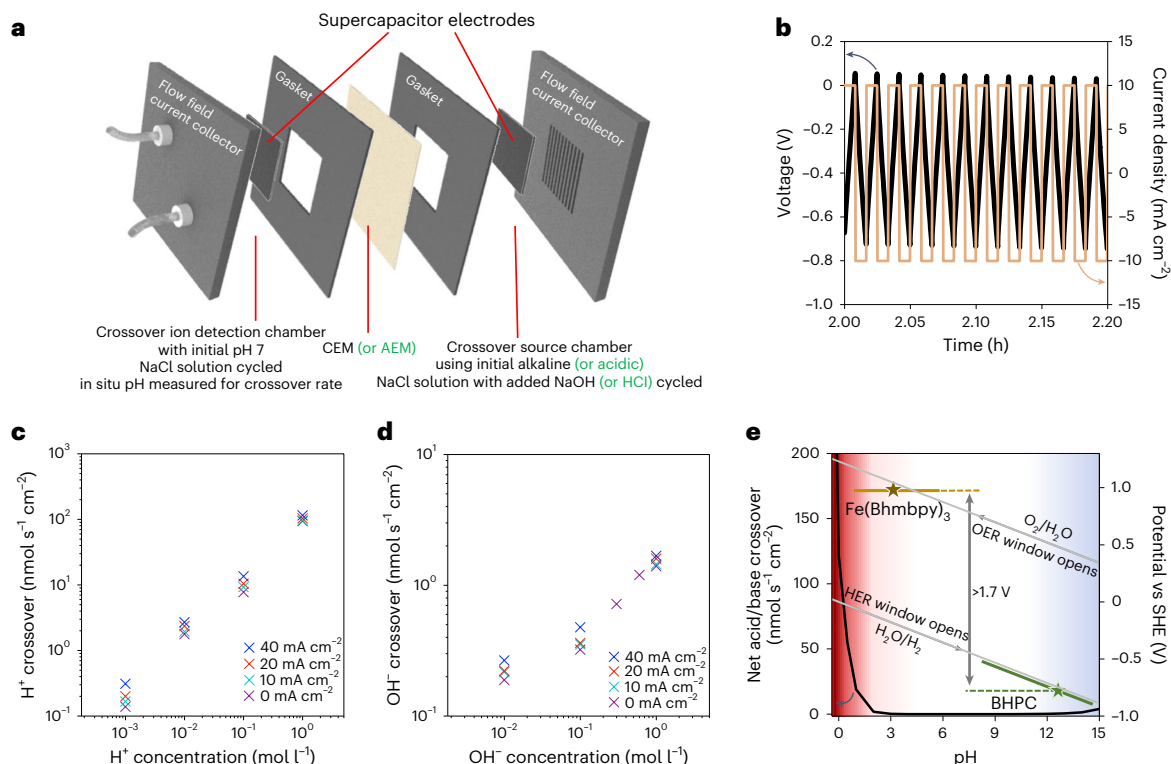


Fig. 2 | Proton/hydroxide crossover flux through AEM/CEM under applied electric field. **a**, Schematic of a supercapacitor cell used to quantify hydroxide and proton crossover. Electrolytes, cell components and sensors are labelled accordingly. Source and acceptor chamber electrolytes are fed by separate tanks. Green text means 'respectively', using the condition in black or green as a set. **b**, Temporal profile of voltage and current density during the crossover supercapacitor test for pH 0 acid source at 10 mA cm⁻². **c**, Proton crossover fluxes through DSV-N as a function of proton concentration in source chamber

for various current densities. **d**, Hydroxide crossover fluxes through E620 as a function of hydroxide concentration in source chamber for various current densities. **e**, Pourbaix diagram with net acid-base crossover flux. The black curve shows the net acid-base crossover flux calculated from **c**, **d**. Background shading is used to indicate the severity of acid-base crossover with DSN-N or E620 (red for acid crossover, blue for base crossover, corresponding to the black crossover flux curve). The thermodynamic water-splitting window is identified by grey lines. Redox potentials of BHPC and Fe(Bhmbpy)₃ are labelled in the diagram.

(DSV-N) as the AEM, for evaluation of crossover fluxes, which were estimated by tracking the pH change of the initially neutral acceptor solution (Supplementary Fig. 1). To determine the effect of proton/hydroxide concentration and applied current density on the crossover fluxes, we varied the initial pH for the source chamber side and applied different alternating current densities in the constructed supercapacitor cell, mimicking the charge-discharge cycles during real battery operation. A typical voltage and current density profile is shown in Fig. 2b. The operation parameters were chosen to avoid water splitting so that the pH change was affected only by crossover. Then the rate of pH change was converted to crossover flux based on solution volume and membrane area. Figure 2c,d shows the measured crossover fluxes for protons and hydroxide, respectively. In both cases, the applied current density has only a minor effect on the magnitude of the crossover fluxes. Conversely, the initial concentration of ions in the source chamber has a substantial, almost linear, impact on the crossover flux. Therefore, we can simplify the net crossover flux into $J_{\text{crossover}} = \max(C_{\text{H}^+}^{\text{AEM}}[\text{H}^+]_{\text{A}}, C_{\text{OH}^-}^{\text{CEM}}[\text{OH}^-]_{\text{B}})$ in which $C_{\text{H}^+}^{\text{AEM}}$ is the crossover coefficient of protons through the AEM and $C_{\text{OH}^-}^{\text{CEM}}$ is the crossover coefficient of hydroxide through the CEM.

By fitting the data of Fig. 2c,d, we found that $C_{\text{H}^+}^{\text{DSV-N}}$ is about 1.7 $\mu\text{m s}^{-1}$ and $C_{\text{OH}^-}^{\text{E620}}$ is about 0.015 $\mu\text{m s}^{-1}$ under zero current. As a result, using these two membranes, concentrated strong acid solutions should be used with particular caution to avoid energy efficiency loss associated with proton crossover in pH-decoupling ARFBs.

We hypothesized that rapid crossover can be prevented while maintaining high cell voltage through 'mild' pH decoupling.

Operation at pH 13 and 3 (a difference of 10 pH units) can still allow an OCV higher than 1.7 V if the negolyte and posolyte are chosen appropriately, as seen in Fig. 2e. By using milder acid and base than previous pH-decoupling batteries, we can still achieve relatively high voltage while exponentially reducing the unwanted acid-base crossover flux ($C_{\text{H}^+}^{\text{DSV-N}}[\text{H}^+]_{\text{A}} \approx 0.1 \text{ nmol s}^{-1} \text{ cm}^{-2}$, $C_{\text{OH}^-}^{\text{E620}}[\text{OH}^-]_{\text{B}} \approx 0.2 \text{ nmol s}^{-1} \text{ cm}^{-2}$, under zero current), thus approaching an optimal balance in the trade-off between operating voltage vs energy inefficiency and instability due to acid-base crossover associated with the membranes we used.

A mild pH-decoupling battery with pH recovery

As a proof of concept, Fe(Bhmbpy)₃ and BHPC were paired in the mild pH-decoupling ARFB. The redox potential of each molecule was first evaluated in a single three-electrode cell (Supplementary Figs. 2 and 3). The redox potential of Fe(Bhmbpy)₃ is pH independent but that of BHPC, which undergoes proton-coupled electron transfer, is pH dependent. Their redox potentials at different pH values are indicated by solid lines in Fig. 2e. Operational conditions for Fe(Bhmbpy)₃ and BHPC were carefully optimized through extensive evaluation (Supplementary Figs. 4–9). In our optimized ARFB, Fe(Bhmbpy)₃ was employed at pH 2–4 and BHPC at pH 13. Furthermore, we chose Fe(Bhmbpy)₃ to be the capacity-limiting side because of its higher fade rate (Supplementary Fig. 10)¹⁸.

Figure 3a,b shows the temporal evolution of the pH of the reservoirs, the cell capacity and the Coulombic efficiency, measured for a test cell in which the negolyte contained 0.1 M Na₂BHPC at pH 13 (20 ml), whereas the posolyte contained 0.1 M Fe(Bhmbpy)₃Cl₂ at pH 3 (10 ml). The supporting electrolyte on both sides was 1.0 M NaCl, with

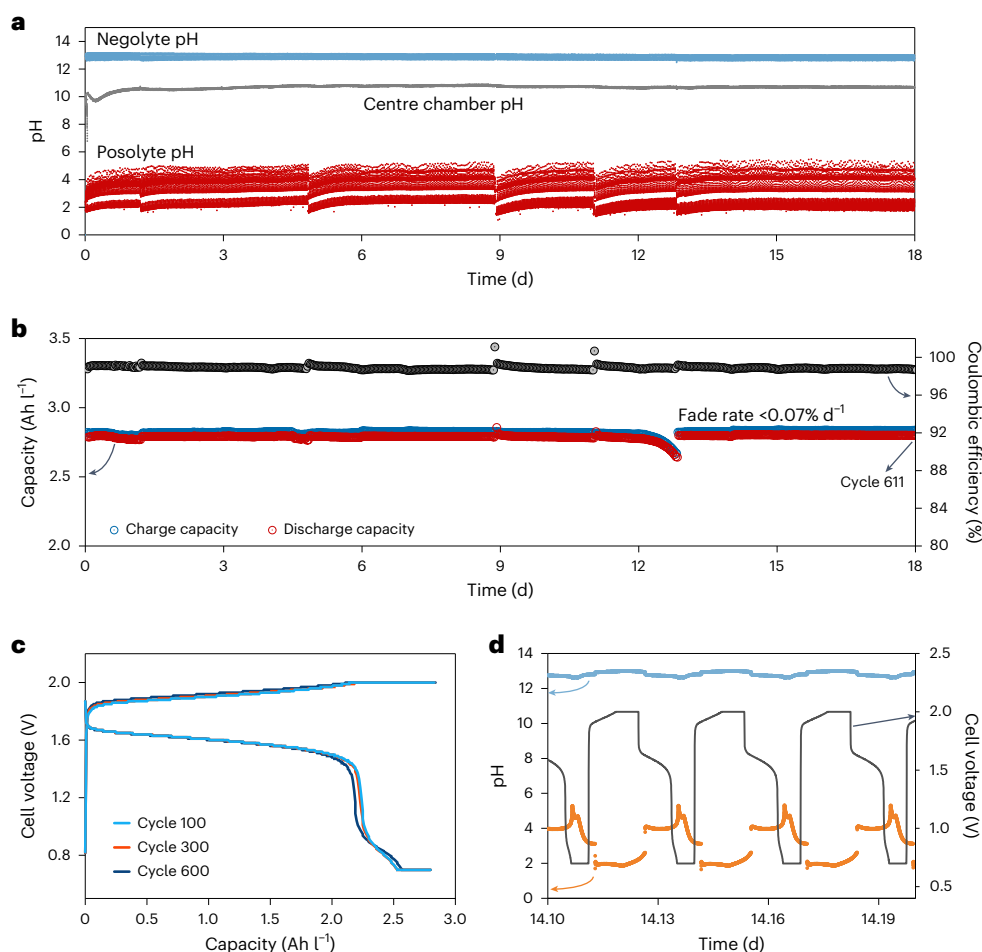


Fig. 3 | Long-term performance of high-voltage mild pH-decoupling battery. **a,b**, Negolyte pH, posolyte pH, capacity and Coulombic efficiency in pH-decoupled ARFB during long-term cycling. The cell was operated with a constant current (20 mA cm⁻²) followed by constant potentials of 2.0 V (charging) and 0.7 V (discharging) during 18 days of operation at room temperature. The cell was assembled with a negolyte comprising 0.1 M BHPC in 1 M NaCl at pH 13 and a

posolyte comprising 0.1 M Fe(Bhmbpy)₃ in 1 M NaCl at pH 3. The middle chamber was circulated with 1.3 M NaCl solution and separated from the side chambers by a CEM and an AEM. **c**, Charge–discharge voltage–capacity profiles from selected cycles. **d**, Zoomed-in temporal profile of cell voltage and pH of negolyte and posolyte during a few cell cycles.

added NaOH or HCl to adjust the pH. Twenty ml 1.3 M NaCl was cycled in the middle chamber. An E620 CEM separated the negolyte from the middle chamber and a DSV-N AEM separated the posolyte from the middle chamber. This cell was cycled with the standard constant current followed by constant voltage (CCCV) protocol at 20 mA cm⁻² with voltage cut-off values of 2.0 V and 0.7 V during charge and discharge, respectively, held until the current density dropped below a current cut-off. The cell accessed >95% of its theoretical capacity and exhibited an average Coulombic efficiency of 99%, presumably limited by the self discharging of the iron-based posolyte¹⁸.

Starting with the simple approach of restoring the initial pH difference through the simultaneous addition of acid and base, 0.1 ml of 1 M HCl was added to the posolyte side, and 0.1 ml of 1 M NaOH was added to the negolyte side when pH had notably drifted from its initial value. The amount and timing of adding exogenous acid–base can be estimated according to the calculated net acid–base crossover fluxes. These addition events are apparent from the discontinuities in the pH data shown in Fig. 3a. The cell was cycled over 18 days, and high Coulombic efficiency and a low capacity fade rate were observed (Fig. 3b). The voltage–capacity profiles of this battery during three representative cycles are shown in Fig. 3c, demonstrating its excellent stability. The plateau near 0.8 V during discharge has been identified as the reversal of a dimerization side reaction of Fe(Bhmbpy)₃¹⁸.

Because the Coulombic efficiency of the system is limited by the posolyte side, the cell drifted from perfect charge balance towards reduction of both sides over roughly 13 days (Fig. 3b, day 13). To counteract this imbalance, we briefly exposed the reduced negolyte to air, which brought the cell back into balance. Negolyte exposure to air is a simple and general strategy to restore out-of-balance ARFBs when a parasitic reduction of the posolyte limits the Coulombic efficiency^{20,21}. We also tested a capacity-balanced cell with the same negolyte and posolyte composition. By exposing negolyte to air intermittently to oxidize it back to balanced charge, we were able to run this capacity-balanced cell with similar slow fading and stable voltage output (Supplementary Fig. 11).

A zoomed-in view of the voltage and pH profile for a few cycles is shown in Fig. 3d. During cell cycling, the pH of the negolyte oscillated due to proton-coupled electron transfer of BHPC. We attribute the oscillating pH of the posolyte to the reversible dimerization side reaction of the oxidized Fe(Bhmbpy)₃¹⁸. The posolyte pH oscillates between 2 and 5, representing a small change in proton concentration (10⁻² mol l⁻¹).

Repeated addition of exogenous acid and base into the reservoirs to maintain the pH difference is not viable in the long term. Over months of cell operation, dissolved salts in the cell will accumulate through the net acid–base crossover reaction: HCl + NaOH = NaCl + H₂O.

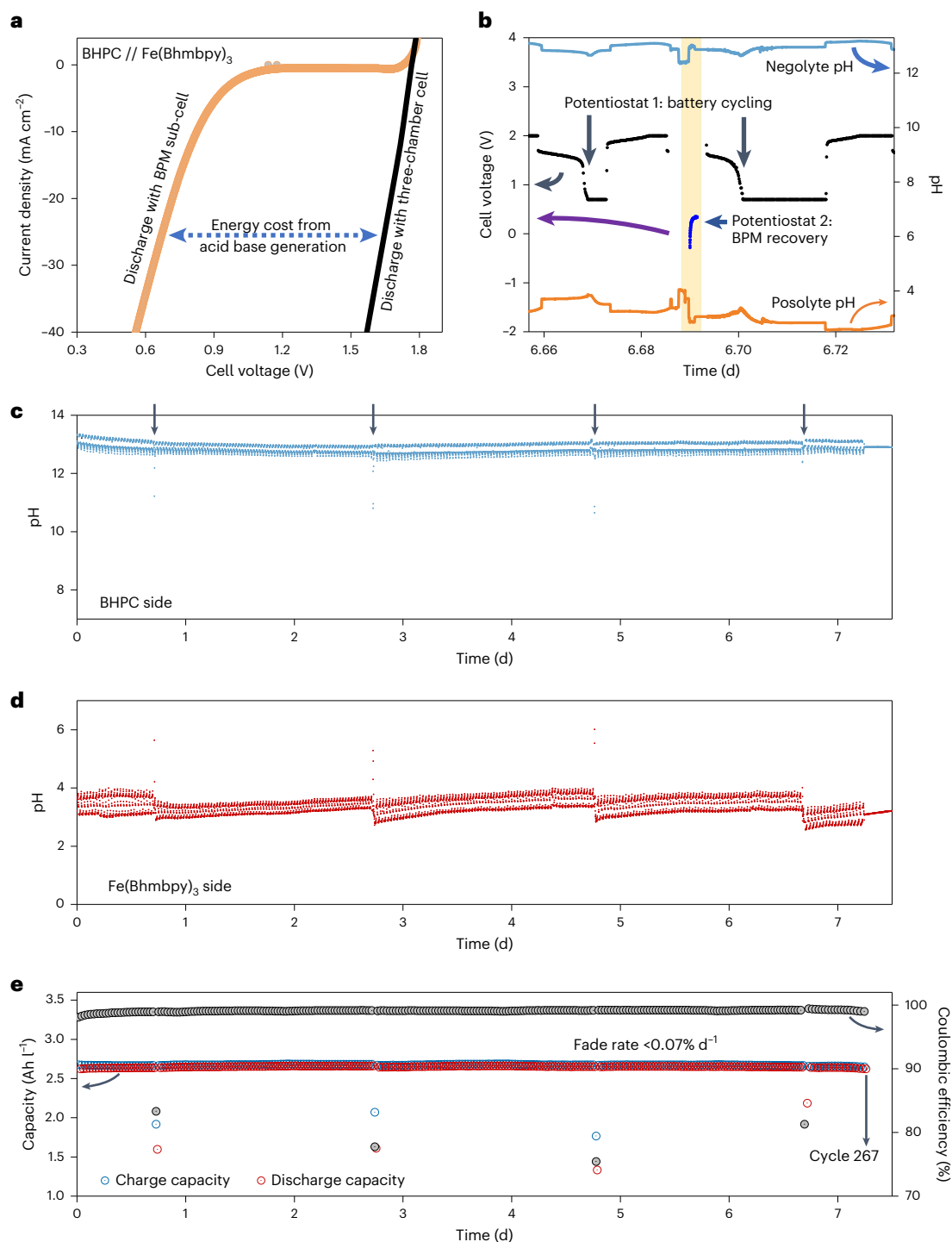


Fig. 4 | Performance of the BPM sub-cell and long-term ARFB cycling with pH recovery. **a**, LSV curve of the BPM sub-cell and three-chamber cell. Electrolytes are pumped into BPM sub-cell at 60–90% state of charge. LSV curve was collected with a sweep rate of 3 mV s^{-1} to discharge the electrolytes. The OCV is about 1.75 V and water dissociation starts around 0.95 V, indicating a voltage cost of around 0.8 V at 5 mA cm^{-2} . **b**, Cell voltage and pH during a few cell cycles and during pH recovery. The CCCV protocol was applied during cell cycling. The BPM sub-cell recovery phase is highlighted in orange. During recovery, potentiostat 1 was disconnected and the BPM sub-cell was independently connected to potentiostat 2. Charged electrolytes from the three-chamber cell were pumped

into the sub-cell. 40 mA cm^{-2} was applied until pumped electrolytes discharged by approximately 6 mAh. In situ pH measurements show pH recovery. **c–e**, Negolyte pH, posolyte pH, capacity and Coulombic efficiency in pH-decoupling ARFB during long-term cycling with pH recovery. Vertical arrows indicate four pH recovery operations. During recovery, electrolytes were pumped through a BPM sub-cell and discharged to dissociate water. The capacity fade rate was smaller than 0.07% per day, and Coulombic efficiency was around 99%. The Coulombic efficiency drop during recovery was due to capacity utilization during BPM sub-cell discharge.

Because these salts are difficult to remove, they will eventually reach saturation and precipitate, leading to cell failure. To avoid this trajectory, we used a BPM sub-cell to recover the pH instability by dissociating

water into protons and hydroxide ions during discharging of the electrolytes. In this way, no exogenous materials are introduced into the system.

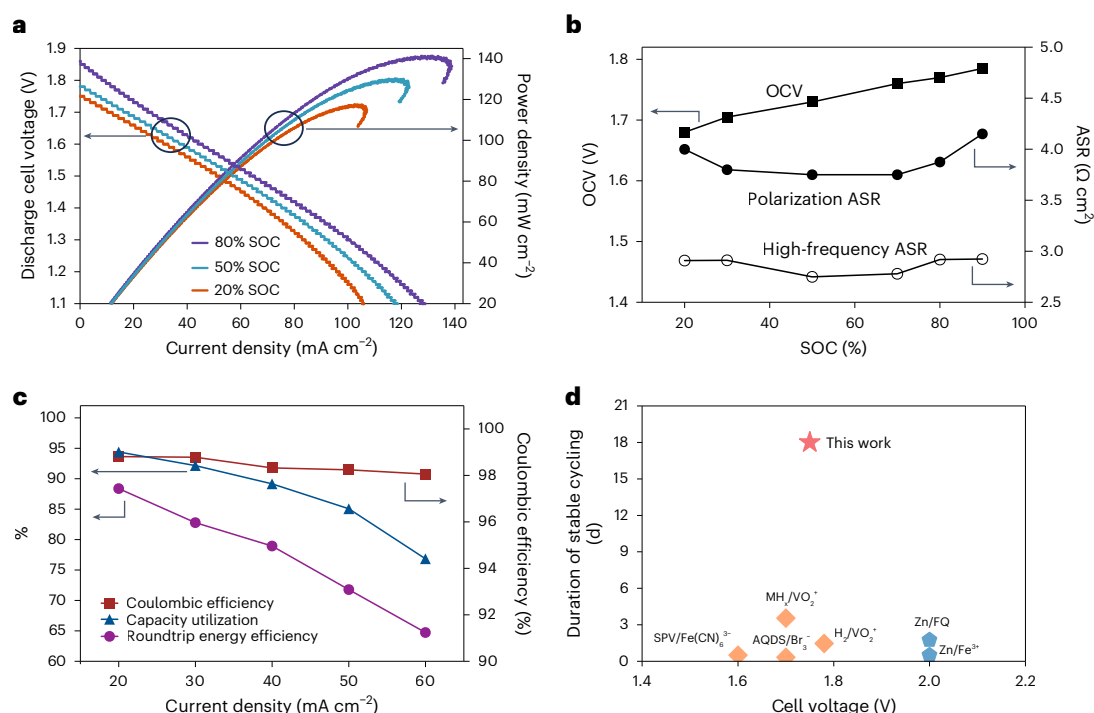


Fig. 5 | Polarization performance of the system and comparison of pH-decoupling systems. **a**, Cell voltage and power density during discharge of three-chamber acid–base ARFB with 0.1 M BHPC in 1 M NaCl at pH 13 (adjusted by NaOH) as negolyte and 0.1 M $\text{Fe}(\text{Bhmbpy})_3$ in 1 M NaCl at pH 3 (adjusted by HCl) as posolyte. The 200- μm thick middle chamber was circulated with 2 M NaCl solution and separated from the side chambers by E620 as CEM and DSV-N as AEM. **b**, Dependence on SOC of OCV and ASR near OCV for the described cell.

c, Capacity utilization and Coulombic and round-trip energy efficiencies vs. applied current density for the cell. **d**, Reported stable cycling duration of previous acid–base ARFBs compared with the current work. Orange points represent BPM pH-decoupling ARFBs. Blue points represent two-membrane pH-decoupling ARFBs. The star represents this work, featuring a long-term practical pH-decoupling ARFB that rebalances the lost acid–base during operation.

To demonstrate the effectiveness of this pH recovery strategy, we pumped partially charged electrolytes through a BPM sub-cell and discharged them across the BPM. Linear-sweep voltammetry (LSV) was applied to study the polarization of the BPM cell (Fig. 4a). In Fig. 4a, we also show the LSV towards discharge in the three-chamber cell. The voltage difference between the two discharge curves is primarily from the energy cost of water dissociation in the BPM sub-cell. The ideal thermodynamic cost of generating 1 M each of acid and base from water is equivalent to 0.83 V, assuming unit activities of protons and hydroxide in their respective membrane layers. The applied voltage for water dissociation, 0.83 V plus associated overpotentials, creates an energy efficiency loss associated with pH recovery. More detailed polarization curves of the BPM using different pH pairs were also recorded in a four-electrode set-up²² (Supplementary Fig. 12).

During pH recovery, we applied 40 mA cm^{-2} constant current density on the BPM cell through an independent potentiostat. A typical BPM cell voltage profile during pH recovery is shown in Fig. 4b (dark blue curve, orange-highlighted region). The cell voltage during a few CCCV cycles before and after recovery is also shown. If the acid–base crossover is rapid, a large amount of charged electrolyte will have to be discharged in the BPM sub-cell, and the resulting energy efficiency penalty will be severe (Supplementary Fig. 13). When the amount of acid–base that must be recovered approaches the amount of charged electrolyte (equivalent acid–base crossover current density exceeds about 10% of the current density for battery cycling), the loss in round-trip energy efficiency of the pH-decoupling ARFB will make it impractical. When the equivalent acid–base crossover current density is small, the round-trip energy inefficiency caused by pH recovery can be negligible, and the geometric area of the BPM sub-cell running under a certain current density can be small enough that it will not excessively increase the capital cost of the battery system.

In our mild pH-decoupling scheme, low acid–base crossover fluxes can be achieved and just four pH recovery cycles (each with a capacity of about 6 mAh) were sufficient to keep the system stable for over a week (Fig. 4c–e, at vertical arrows). After each recovery cycle, the pH of the negolyte increased (Fig. 4b, light blue trace, and Fig. 4c), whereas the pH of the posolyte decreased (Fig. 4b, orange trace, and Fig. 4d). The voltage–capacity profile shows stable cycling after pH recovery (Supplementary Fig. 14). The resistance of the BPM increased after two weeks of usage; we attribute this to the degradation of the catalytic layer in the BPM. Next-generation BPMs with lower resistance, lower water dissociation overpotential and higher stability can further improve this system.

It is worth noting that the pH-difference recovery does not necessarily have to be periodic. pH recovery can be operando and conducted under steady state by discharging the electrolyte in the BPM sub-cell at a constant current matching the equivalent acid–base crossover current. In this case, the energy cost from pH recovery presents as a Coulombic inefficiency during the cycling of the main ARFB.

To increase energy efficiency and power density of the system, we further optimized the cell structure. The resistance of the three-chamber cell was first studied by varying the thickness of the centre chamber (Supplementary Figs. 15 and 16). To achieve a sub-millimetre centre chamber thickness, we kept the AEM and CEM separated by only 200 μm with a porous spacer in between (Supplementary Fig. 17). Ohmic resistance is linearly decreased by a reduction in the centre chamber thickness (Supplementary Fig. 18). Centre-chamber NaCl concentration, temperature and flow rate were also investigated and optimized (Supplementary Fig. 19–23). With these modifications to the centre chamber and cycling conditions, the high-frequency ohmic resistance of the system was reduced to nearly the sum of two membrane resistances (Supplementary Fig. 24).

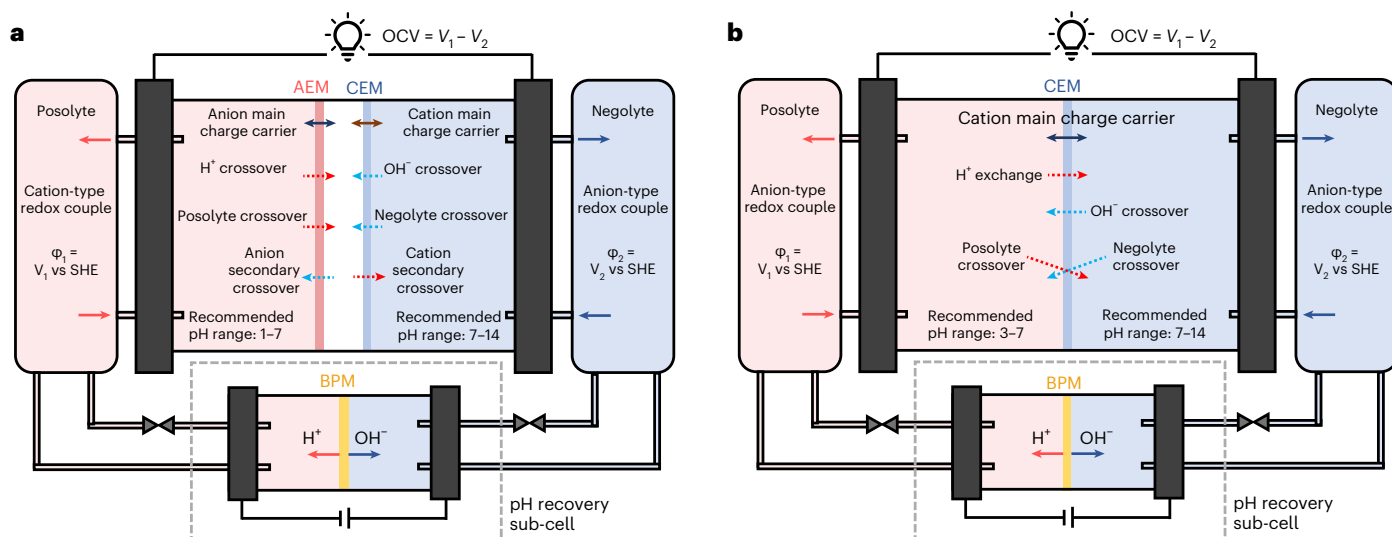


Fig. 6 | pH-decoupling ARFBs with pH recovery. a, A three-chamber, two-membrane pH-decoupling ARFB with pH recovery. The cell can use concentrated acid and base, while considering the cost of acid–base crossover. **b**, A CEM-type single-membrane pH-decoupling ARFB with recovery. The acidic side has to be mild for the main charge carrier to be non-proton

cations. Correspondingly, an AEM-type single-membrane pH-decoupling ARFB should use mild alkaline electrolyte against acidic electrolyte for the main charge carrier to be non-hydroxide anions. The OCV of the cell is determined by the electrode potential of the posolyte (V_1) and that of the negolyte (V_2).

We then conducted a galvanostatic polarization analysis of the optimized cell structure with a negolyte composition of 0.1 M BHPC at pH 13 (10 ml) and a posolyte composition of 0.1 M Fe(Bhmbpy)₃ at pH 3 (4.5 ml). The potential was held at the end of each discharge to recover Fe(Bhmbpy)₃ from dimerization. Polarization curves and galvanic power densities are shown in Fig. 5a. The three-chamber pH-decoupling battery reached a peak power density of 140 mW cm⁻² at 80% state of charge (SOC). Figure 5b shows the OCV of the cell, increasing from 1.69 V at 20% SOC to 1.8 V at 90% SOC. The high-frequency area-specific resistance (ASR) of the battery averaged 2.85 Ω cm² whereas the average polarization ASR across all SOC was 3.82 Ω cm². These values indicate that most of the cell's overall resistance is due to the resistance of the membranes. Supplementary Fig. 25 depicts the voltage profile of the cell at different current densities from which we extracted the Coulombic efficiency, capacity utilization and round-trip energy efficiency for Fig. 5c. This high-voltage cell exhibits 80% round-trip energy efficiency at 40 mA cm⁻².

With this mild pH-decoupling set-up with pH recovery, exhibiting a crossover flux smaller than 0.3 nmol s⁻¹ cm⁻², we have a negligible energy and capital cost penalty due to the recovery system. Thus, we have realized a long-term practical pH-decoupling ARFB, theoretically permitting an indefinite duration of operation and demonstrating longer operation with slower fading than previous examples^{7,8,10,19,23,24} as illustrated by the duration of stable cycling reported in Fig. 5d. We report this to emphasize that future pH-decoupling cell designs should place a greater focus on long-term operation and techno-economic considerations²⁵. To these ends, higher voltage output and lower ASR can both be beneficial by lowering the relative cost of pH recovery, thereby making higher acid–base crossover rates tolerable.

Other applications and techno-economic calculations

The application of our cell design and recovery concept is not limited to one redox couple pair. To illustrate the generalization of this system to other redox chemistries, we demonstrated cycling of three other pairs of electrolytes in mild pH-decoupling ARFBs. Fe(Bhmbpy)₃ at pH 3 and di-butanoate ether anthraquinone (DBEAQ)²⁶ at pH 12 were used to demonstrate the cell performance under milder pH difference with non-PCET electrochemistry in both sides (Supplementary Figs. 26

and 27). This milder pH-decoupling three-chamber cell exhibited better stability and required infrequent addition of acid and base. With pH recovery, the cell also showed similar BPM overpotential to that of the model cell with BHPC, and we measured an overall fade rate of <0.07% per day with 99% Coulombic efficiency (Supplementary Figs. 28–30). The polarization test of this cell revealed similar high energy efficiency but a slightly smaller power density because of its lower OCV (Supplementary Fig. 31).

By exchanging the DBEAQ negolyte for Fe(DIPSO)^{27,28}, an iron complex negolyte species that operates at pH about 13, we achieved a pH-decoupled all-iron ARFB with OCV around 1.8 V (Supplementary Figs. 32–36). Maintaining high pH is essential for the stability of Fe(DIPSO), as it precipitates from solution at lower pH. Long-term cycling with either exogenous acid–base addition or pH recovery exhibited a low fade rate and low pH recovery cost.

Further, we utilized a high concentration aqueous redox pair: 2 M FeCl₃ (pH 1–3) was paired with 1 M BHPC (pH 13–14, introducing high pH swing from the PCET electrochemistry) realizing high volumetric capacity. The results demonstrated promising cycling performance and successful pH recovery through the BPM cell (Supplementary Fig. 37). However, the results also showed fast capacity fading due to iron ion crossover. Towards the practical application of pH-decoupling ARFB, the crossover of redox-active molecules has to be controlled. Researchers have developed larger molecules²⁹, higher charged molecules or better ion-selective membranes with stronger repulsion against redox-active molecules^{30–34} to suppress crossover. Another strategy is to develop ARFB systems allowing a mixture of both high-potential and low-potential redox species in both posolyte and negolyte so that crossover is inconsequential³⁵.

For higher voltage, pairing Fe(Bhmbpy)₃ at pH 3 with Cr chelate 1,3-propylenediaminetetraacetic acid (CrPDTA) at pH 9 can reach an OCV about 2.2 V (ref. 6), with net acid–base crossover flux lower than 0.2 nmol s⁻¹ cm⁻². Other suitable redox candidates are listed in Supplementary Table 1, from which we can find redox pairs for operation of pH-decoupling design with various values of OCV and crossover flux.

To gain insight into the future development of such pH-decoupling systems, we studied the cost penalty from acid–base crossover by doing a techno-economic analysis of the penalties that are inherent to any pH-decoupled system (Supplementary Fig. 38)^{25,36}. We assumed that

the BPM sub-cell operates at a steady-state current density (equal to the equivalent acid–base crossover current) for pH recovery. Consequently, higher acid–base crossover flux leads to higher capital cost associated with a larger BPM sub-cell. This also results in higher energy cost for pH recovery, penalizing the round-trip energy efficiency. Acid–base crossover will be less impactful when using a battery stack with smaller membrane area relative to the volume of electrolyte tanks. Because the required geometric battery area for a certain power output decreases roughly as $(\text{ASR}) \times (\text{OCV})^{-2}$, an increased OCV cuts the rate of acid–base crossover for a flow battery system with fixed discharge duration and thereby cuts the BPM cell cost and the energy efficiency penalty of the recovery system. The capital cost of the three-chamber cell was estimated using a reported techno-economic model^{4,37}, reckoning the additional costs associated with a three-chamber two-membrane set-up and a pH recovery system (Supplementary Fig. 39). We estimated the current density of the ARFB as $\frac{0.2 \text{ OCV}}{\text{ASR}}$. The costs of the recovery system can be acceptable if the increased OCV more than offsets the increased ASR associated with an additional membrane and an additional electrolyte chamber.

Because power output and overall ASR appear to be essentially limited by the resistances of the two membranes and because the BPM sub-cell and the energy cost penalty of pH recovery is determined by the acid–base crossover flux through the two membranes, further development of lower-resistance AEMs and CEMs with low acid–base crossover coefficient is important for pH-decoupling ARFBs^{34,38}. We show some of the commercially available AEMs permit very low proton crossover flux ($<1 \text{ nmol s}^{-1} \text{ cm}^{-2}$) even working against pH 0 (Supplementary Fig. 40), which is of great promise for future applications in severe pH-decoupling systems.

Although the high capital cost and the high ASR associated with two membranes challenge the commercialization of three-chamber ARFBs (Supplementary Fig. 41), the implementation of mild pH decoupling can still be useful in other cell designs^{6,20}. For other pH-decoupling systems, which always suffer to some extent from acid–base crossover during long-term operation, pH recovery can also be realized^{45,39}. Even a small pH difference between the cathode and the anode can enlarge the space for utilization of redox-active molecules in high-voltage cells. We constructed a single-membrane two-chamber pH-decoupling ARFB that uses KBr/KBr₃ (1 M, at pH 2–5) against KCrPDTA (1 M, at pH 9–10, buffered by 0.1 M PDTA), separated by only one cation exchange membrane. Because the concentration of K⁺ is at least two orders of magnitude higher than the concentration of H⁺, the charge carrier through the membrane is predominantly K⁺. Meanwhile, hydroxide anions are mostly rejected by the CEM by charge exclusion. Therefore, the net acid–base crossover flux in the system is low, which enables the system pH gradient to be relatively stable. Single-membrane pH-decoupling ARFBs have higher potential for commercialization and higher tolerance for acid–base crossover flux due to their increased OCV without extra membranes or chambers compared to conventional ARFBs (Supplementary Fig. 42). The high-frequency ASR of the cell we assembled is about $1.2 \Omega \text{ cm}^2$, with an open circuit voltage around 2.1 V. With this cell set-up, we achieved a peak power density higher than 600 mW cm^{-2} (Supplementary Fig. 43). We cycled the high concentration (26 Ah l^{-1}) cell, achieving a Coulombic efficiency of approximately 98%, limited by Br₂ crossover (Supplementary Fig. 44). The crossover of Br₂ did not cause decomposition, precipitation or cell failure (Supplementary Fig. 45). We also confirmed that protons and hydroxide ions can be generated by discharging in the BPM sub-cell to initiate the pH difference. Polarization of charged electrolyte through the BPM sub-cell shows a voltage cost of about 1 V for acid–base generation at a current density of 40 mA cm^{-2} (Supplementary Fig. 46). A comparison between two-membrane pH-decoupling ARFB and single-membrane pH-decoupling ARFB is provided in Fig. 6, with recommended operating conditions.

Conclusions

We presented a mild pH-decoupling ARFB, operating with a negolyte pH of -13 and a posolyte pH of -3, suppressing acid–base crossover rates to less than $0.3 \text{ nmol s}^{-1} \text{ cm}^{-2}$. This strategy permits an open circuit voltage exceeding 1.7 V with long lifetime and only minor energy cost and Coulombic efficiency penalty. Additionally, we developed a sub-cell using a bipolar membrane that can recover protons and hydroxide crossover without requiring addition of exogenous acid and base, thereby preventing the long-term salt accumulation problem. This work resolves the long-standing trade-off between high-voltage output and the energy cost penalty associated with acid–base crossover, by answering the question of ‘in pH-decoupling systems, how bad acid–base crossover is and how to manage it’. We also demonstrated that the concept of mild pH decoupling and acid–base recovery can be utilized in other pH-decoupling, pH-sensitive or pH-unstable electrochemical applications. Future work on pH-decoupling batteries should not only pursue high-voltage output but also pay more attention to the long-term practicability of the set-up and the cost associated with maintaining the pH difference. Regarding the capital cost of energy storage, higher voltage output and lower resistance of the pH-decoupling set-up can increase the acid–base crossover tolerance, which informs directions of developing next-generation aqueous redox-active molecules, low-resistance membranes with high selectivity and long-lifetime pH-decoupling ARFB systems.

Methods

Materials and synthesis

4,4'-bis(hydroxymethyl)-2,2'-bipyridine (Bhmbpy), 3-(bis(2-hydroxyethyl) amino)-2-hydroxypropanesulfonic acid (DIPSO), 1,3-propylenediaminetetraacetate (PDTA), FeCl₂, FeCl₃ were purchased from VWR International. 4,4'-((9,10-Dioxo-9,10-dihydroanthracene-2,6-diyl)bis(oxy))dibutyric acid (2,6-DBEAQ) was purchased directly from TCI Chemicals. Cr(II)-(1,3-propylenediaminetetraacetate) potassium salt (CrPDTA) was provided by Marshak group from University of Colorado Boulder. 3,4-diaminobenzoic acid, 2-hydroxy-naphthquinone, potassium ferricyanide and potassium ferrocyanide were purchased from Sigma-Aldrich. Fumasep E-620K and Nafion 117 as cation exchange membrane, Selemion DSV-N anion exchange membrane and Fumasep bipolar membrane were purchased and only soaked in 1 M NaCl or KCl solution before usage.

Supercapacitor electrodes were fabricated as follows: high surface area porous carbon powder (purchased from ACS Materials), polyvinylidene fluoride (PVDF) (as a binder, dissolved in *N*-methyl-2-pyrrolidone (NMP) with 1:40 PVDF/NMP weight ratio) and SuperP carbon black were combined in a slurry with the following weight ratios: 85:10:5 to create the capacitive electrodes. The slurry was cast on baked carbon papers (SGL 39AA) (400 °C overnight) before it was dried overnight at 70 °C. The electrodes were then used directly for the crossover tests in a standard flow cell.

Electrodes for cell cycling tests are carbon papers (SGL 39AA) baked at 400 °C overnight. Centre chambers for three-chamber cells were made by PDMS moulding or 3D printing (Supplementary Information: ‘Three-chamber two-membrane cell set-up characterization and optimization’). The centre chamber thickness varies from 0.2 mm to centimetres. Several porous plastic spacers were put in the centre chamber window to physically avoid AEM/CEM touching each other. For this paper, we were using a 3D printed centre chamber with curved-in membrane to achieve even flow with 0.2 mm thickness for cell cycling tests, if not further specified.

For BHPC synthesis, 10 mmol of 3,4-diaminobenzoic acid in 100 ml glacial acetic acid, with 10 mmol 2-hydroxy-naphthquinone slowly added was stirred for 12 h at 50 °C. BHPC was obtained with yield >90% after pouring the mixture into cold water and washing. BHPC electrolyte was prepared by dissolving a certain concentration of BHPC in 1 M NaCl, with NaOH adjusting the pH to 13.

$\text{Fe}(\text{Bhmbpy})_3$ electrolyte was prepared by dissolving FeCl_2 and Bhmbpy with a ratio of 1:3.5 in 1 M NaCl, using HCl adjusting pH to 3, if not claimed otherwise. FeCl_2 electrolyte was prepared by dissolving 2 M FeCl_2 in water, with NaCl and HCl adjusting the whole ion concentration and initial pH to 2.

$\text{Fe}(\text{DIPSO})$ electrolyte was prepared by dissolving FeCl_3 and DIPSO with a ratio of 1:1.5, with NaCl and NaOH adjusting the whole ion concentration (total anion equalling 1.3 M) and initial pH to 13. The electrolyte was filtered before usage. DBEAQ electrolyte was prepared by dissolving 0.1 M DBEAQ in 1 M KCl, with KOH adjusting the pH to 12. When using DBEAQ, 1.3 M KCl was used instead of NaCl in the centre chamber when running the three-chamber cell.

CrPDTA electrolyte was prepared by dissolving 1 M CrPDTA (potassium salt) in water, then adding 0.1 M PDTA as buffer reagent and adjusting pH to 9 using KOH.

Crossover of hydroxide ions and protons

The crossover flux was evaluated in a standard 5 cm² flow cell with two chambers (receiving and source chambers). The receiver chamber contains 1 M NaCl, and its initial pH was set at 7. The source chamber contains 1 M NaCl, and its pH was varied by NaOH (or HCl) for each crossover test (1 M NaOH for pH 14 and 1 M HCl for pH 0, without NaCl). Each cell was built with two layers of high surface area porous carbon-coated supercapacitor electrodes in the anode and the cathode that are separated by an ion exchange membrane (DSV-N for proton crossover test and E620 for hydroxide crossover test). The membrane was soaked in 1 M NaCl overnight before use. During each crossover test, alternating constant current was applied with a restricted cell voltage of 1.1 V to avoid water splitting. During alternating current cycling, the pH of both the source and receiving chambers were tracked by pH sensors in real time.

Cyclic voltammetry

This was recorded by an electrochemical analyser from CH instruments CHI 608E with a three-electrode system including one glassy carbon working electrode, one platinum counter electrode and one Ag/AgCl reference electrode. The glassy carbon working electrode was polished with an aluminium slurry before and after each scan.

Rotating disk electrode

Tests were performed using a Pine Instruments Modulated Speed Rotator AFMSRCE equipped with a 5 mm diameter glassy carbon working electrode (Pine Instruments ESPK), an Ag/AgCl reference electrode in 3 M KCl solution (CHI), a graphite counter electrode and a Gamry Reference 3000 potentiostat.

Three-chamber, two-membrane cell cycling

This was conducted in a 5 cm² flow battery, using 10 ml 0.1 M $\text{Fe}(\text{Bhmbpy})_3$ as capacity-limiting side, pairing with 20 ml 0.1 M negolyte, if not clarified. Electrolytes were pumped using a flow rate of 60 ml min⁻¹ through three layers of based carbon papers as electrodes. Centre chamber is separated by one E620 and one DSV-N. Twenty ml 1.3 M NaCl was cycled as centre chamber electrolyte. Centre solution was pumped using 10 ml min⁻¹. Three pH sensors were inserted in posolyte, negolyte and centre solution, respectively. pH reading out was collected by an Arduino during cell operation and manually calibrated after. The cell was operated with a constant current (20 mA cm⁻²) followed by constant potentials of 2 V (charging) and 0.7 V (discharging). During charging, the potential was held until the current density dropped lower than 1 mA cm⁻². During discharge, the current density cut-off was 0.2 mA cm⁻² to ensure full discharge of the iron complex and its dimers. During operation, we would manually add 0.1 ml 1 M HCl and NaOH to compensate the acid–base crossover when the posolyte pH reaches 5. Or in pH recovery experiments, we would conduct BPM sub-cell pH recovery.

BPM sub-cell pH recovery

Recovery was conducted in a 2.5 cm² flow battery with a BPM sandwiched in two layers of baked SGL-AA carbon paper. The sub-cell electrolyte inlets and outlets shared the electrolyte tank of the three-chamber cell. The sub-cell was connected to an independent potentiostat. During recovery, the cell cycling of the three-chamber cell was paused. Then we opened the valve and started the pump connecting sub-cell, with partially charged electrolytes (> 50% SOC) pumped through. Electrolytes are discharged in sub-cell using a constant current of 40 mA cm⁻² for 3–4 min to recover the pH gradient loss. Afterwards, electrolytes were pumped out of the sub-cell, with valve closed and three-chamber cell normal cycling restarted. BPM polarization of certain pairs of electrolytes were evaluated by LSV with a sweeping rate of 3 mV s⁻¹, scanning from a voltage close to OCV to lower voltage.

Single-membrane pH-decoupling cell cycling

This was conducted in a 5 cm² flow battery, using 5 ml 1 M CrPDTA as capacity-limiting side, pairing with 10 ml 1 M KBr. Three layers of baked carbon papers were separated by one Nafion 212. Electrolytes were pumped using a flow rate of 60 ml min⁻¹. Two pH sensors were inserted in posolyte and negolyte, respectively. pH reading out was collected by an Arduino during cell operation and manually calibrated after. The cell was operated with a constant current (50 mA cm⁻²) followed by constant potentials of 2.4 V (charging) and 1.4 V (discharging). The potential was held until the current density dropped lower than 20 mA cm⁻². We conducted BPM sub-cell acid–base generation before the starting of cell cycling. The pH in CrPDTA should be between 9 and 10, whereas KBr should be lower than 7 to ensure the normal cycling.

Cell polarization

This was conducted for $\text{Fe}(\text{Bhmbpy})_3$ -BHPC and $\text{Fe}(\text{Bhmbpy})_3$ -DBEAQ cells.

For the $\text{Fe}(\text{Bhmbpy})_3$ -BHPC test, a cell composed of ~4.0 ml 0.1 M $\text{Fe}(\text{Bhmbpy})_3$ (10.7 mAh theoretical capacity) in 1 M NaCl at pH 3 (HCl was added to adjust pH) paired with 5 ml of 0.1 M BHPC (26.8 mAh theoretical capacity) in 1 M NaCl at pH 13 (NaOH was added to adjust pH). The middle chamber was circulated with 2 M NaCl solution sandwiched by E620 as CEM and DSV-N as AEM. Three layers of carbon papers (SGL 39AA) baked at 400 °C overnight were used in each half cell. For the $\text{Fe}(\text{Bhmbpy})_3$ -DBEAQ cell, conditions were similar, but supporting electrolytes were changed into KCl.

The full accessed capacity of the cell was first determined by performing a full charge/discharge cycling, which was accomplished by applying a 20 mA cm⁻² constant current followed by potential holds at 2 V (charge) and 0.7 V (discharge) until the current decreased to the cut-off value. The cell was then charged with intervals to various states of charge (calculated from the percentage of the total accessed capacity). To determine the high-frequency area-specific resistance (ASR), electrochemical impedance spectroscopy measurements were conducted at each SOC with a perturbation of 10 mV and frequencies ranging from 1 to 100,000 Hz. The battery's open circuit voltage (OCV) was also measured. Additionally, the potential was swept at each SOC at a scan rate of 150 mV s⁻¹, and the current response as a result was measured to create polarization curves. The polarization ASR was computed using the linear region of the polarization curves near OCV.

For galvanostatic tests, constant current densities between 10 to 60 mA cm⁻² with intervals of 10 mA cm⁻² were applied during both charge and discharge cycles with potential holds during discharge. Cut-off values of 2.2 V and 0.7 V were used. Consequently, the Coulombic, capacity utilization and energy efficiencies are calculated for each current density.

Data availability

The datasets analysed and generated during the current study are included in the paper, its Supplementary Information and uploaded to Figshare at <https://doi.org/10.6084/m9.figshare.23816796>.

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Author contributions

D.X. and A.M.A. designed and conducted cell tests and electrochemical experiments. D.X. developed the concept of mild pH-decoupling ARFB and acid–base recovery. A.M.A. worked on molecule synthesis and characterization. D.X. and J.G. worked on the posolyte. D.X. and T.C. worked on the design of centre chambers. A.M.A., L.C.I.F., T.Y.G. and T.W. worked on crossover characterization. D.X. and Z.Y. worked on the single-membrane pH-decoupling cell. D.X. did the techno-economic calculation. R.Y.L. and R.G.G. supervised the molecular synthesis and characterization. M.J.A. supervised the project. D.X., A.M.A. and M.J.A. drafted the manuscript. All authors edited the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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